Communication with MPU9250 via SPI Interface

Overview

The MPU9250 communicates with the FPGA using the SPI protocol. The device supports SPI Mode 0, which corresponds to:

- **CPOL = 0**: Clock is low when idle.
- **CPHA = 0**: Data is captured on the **rising edge** and propagated on the **falling edge** of the clock.

SPI works in full-duplex mode, allowing simultaneous data transmission and reception.

SPI Signal Descriptions

Signal	Description
SCLK	Serial Clock, driven by FPGA
MOSI	Master Out Slave In (FPGA → MPU9250)
MISO	Master In Slave Out (MPU9250 → FPGA)
CS	Chip Select (Active Low). Set LOW to start communication

> **Note:** MSB is always transmitted first.

SPI Communication Operations

SPI Read Operation

1. Set `CS = 0` (Start communication)
2. Send register address (MSB = 1 to indicate read operation)
3. Send dummy byte (`0x00`) while receiving the response
4. Set `CS = 1` (End communication)

Example:

Reading from WHO_AM_I register:

- **MOSI (FPGA → MPU)**:
    1. `0x75` (register address with MSB = 1)
    2. `0x00` (dummy byte)
- **MISO (MPU → FPGA)**:
    1. `0x80` (previous byte)
    2. `0x71` (response)

SPI Write Operation

1. Set `CS = 0`
2. Send the register address with MSB = 0 (write operation)
3. Send the data byte to be written
4. Set `CS = 1`

Example:

Writing to `PWR_MGMT_1` register (`0x6B`):

- **MOSI (FPGA → MPU)**:
    1. `0x6B`
    2. `0x01` (Power Down)
- **MISO (MPU → FPGA)**:
    1. `0x00` (default previous data)

FPGA-Sensor Communication Example

I/O Description

- **Input:**
  - `clk`: Clock
  - `reset`: Reset signal
  - `[7:0] tx_data`: Transmit data (8-bit)
  - `miso`: Master In Slave Out
- **Output:**
  - `mosi`: Master Out Slave In
  - `sclk`: SPI Clock
  - `cs`: Chip Select
  - `[7:0] rx_data`: Received data (8-bit)
  - `busy`: Communication status flag

Data Transfer Example:

- **FPGA to sensor (via MOSI)**: `0x75` → `01110101`
- **Sensor to FPGA (via MISO)**: `0x3C` → `00111100`

Operation Summary:

- Slave sends `0x3C` via `MISO`, bit by bit on each clock edge.
- Master sends `0x75` via `MOSI`.
- Transfer occurs simultaneously.
- Observed output: `MISO` (sensor output) = `rx_data`

- Given input: `MOSI` = `tx_data`